PGCP-AC AUGUST 26
Database Assignment 1

Schema Design & Table Creation (DDL & DML Commands)

1. Define a schema for a Library Management System with the following entities:
- **Books**
- **Authors**
- **Members**
- **Borrow_Records**

```
mysql> create database Library_Management_System;

Query OK, 1 row affected (0.01 sec)

mysql> show databases;

+----------------------------+

| Database                   |

+----------------------------+

| aug26                      |

| information_schema         |

| library_management_system |

| mysql                      |

4 rows in set (0.00 sec)

mysql> use Library_Management_System;

Database changed

mysql> create table Authors(author_id int primary key, name varchar(100), country varchar(50));

mysql> create table Books(book_id int primary key, title varchar(150), author_id int,
published_year year,      available_copies int,

    -> foreign key(author_id) references Authors(author_id));
```

CDAC MUMBAI

```
mysql> create table Members(member_id int primary key, name varchar(150), email
varchar(100) unique, phone varchar(15));

mysql> create table Borrow_Records(record_id int primary key, member_id int, book_id int,
borrow_date Date, return_date Date,

    -> foreign key(member_id) references Members(member_id),

    -> foreign key(book_id) references Books(book_id));
```

2. Write the SQL command to create a table Authors with the following fields:
- **author_id (Primary Key, INT)**
- **name (VARCHAR(100))**
- **country (VARCHAR(50))**

```
mysql> create table Authors(author_id int primary key, name varchar(100), country varchar(50));

Query OK, 0 rows affected (0.02 sec)
```

3. Write the SQL command to create a table Books with the following fields:
- **book_id (Primary Key, INT)**
- **title (VARCHAR(150))**
- **author_id (Foreign Key referencing Authors)**
- **published_year (YEAR)**
- **available_copies (INT)**

```
mysql> create table Books(book_id int primary key, title varchar(150), author_id int,
published_year year,      available_copies int,

    -> foreign key(author_id) references Authors(author_id));

Query OK, 0 rows affected (0.04 sec)


mysql> desc books;

+-----------------+--------------+------+-----+---------+-------+

| Field           | Type         | Null | Key | Default | Extra |

+-----------------+--------------+------+-----+---------+-------+

| book_id         | int          | NO   | PRI | NULL    |       |

| title           | varchar(150) | YES  |     | NULL    |       |
```

CDAC MUMBAI

| author_id | int | YES | MUL | NULL | |

| published_year | year | YES | | NULL | |

| available_copies | int | YES | | NULL | |

+------------------+-------------+------+-----+---------+-------+

5 rows in set (0.01 sec)

4. Write the SQL command to create a table Members with:
- **member_id (Primary Key, INT)**
- **name (VARCHAR(100))**
- **email (VARCHAR(100), unique)**
- **phone (VARCHAR(15))**

mysql> create table Members(member_id int primary key, name varchar(150), email varchar(100) unique, phone varchar(15));

Query OK, 0 rows affected (0.03 sec)

5. Write the SQL command to create a table Borrow_Records with:
- **record_id (Primary Key, INT)**
- **member_id (Foreign Key referencing Members)**
- **book_id (Foreign Key referencing Books)**
- **borrow_date (DATE)**
- **return_date (DATE)**

mysql> create table Borrow_Records(record_id int primary key, member_id int, book_id int, borrow_date Date, return_date Date,

-> foreign key(member_id) references Members(member_id),

-> foreign key(book_id) references Books(book_id));

Query OK, 0 rows affected (0.05 sec)

6. Modify the Books table to add a column genre of type VARCHAR(50).

mysql> alter table Books

-> add column genre varchar(50);

Query OK, 0 rows affected (0.03 sec)

CDAC MUMBAI

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> select * from Books;
+---------+-----------+-----------+----------------+------------------+-------+
| book_id | title     | author_id | published_year | available_copies | genre |
+---------+-----------+-----------+----------------+------------------+-------+
|     101 | Gitanjali |       123 |           1961 |              100 | NULL  |
+---------+-----------+-----------+----------------+------------------+-------+
1 row in set (0.00 sec)
```

7. Write the SQL command to drop the Borrow_Records table.

```
mysql> Drop table Borrow_Records;
Query OK, 0 rows affected (0.02 sec)
mysql> show tables;
+------------------------------+
| Tables_in_aug26              |
+------------------------------+
| authors                      |
| books                        |
| course                       |
| department                   |
| employee                     |
| members                      |
| passport_application         |
| passport_application1        |
| passport_application_details |
```

CDAC MUMBAI

| passport_registration |

| student |

| student1 |

| student2 |

| student_backup |

| student_course |

+------------------------------+

15 rows in set (0.00 sec)

8. Insert 3 records into the Authors table.

mysql> insert into Authors(author_id, name, country) values (123, 'Shakesphere', 'America'),(456, 'Rabindranath','India'),(789, 'Vikram', 'India');

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> select

   -> * from Authors;

+-----------+--------------+---------+

| author_id | name | country |

+-----------+--------------+---------+

| 123 | Shakesphere | America |

| 456 | Rabindranath | India |

| 789 | Vikram | India |

+-----------+--------------+---------+

9. Insert 5 books into the Books table.

mysql> insert into Books(book_id, title, author_id, published_year, available_copies) values(100,'Gitanjali', 123, 1961, 100), (102,'Once Upon a Time', 456,1962, 1500), (103,'Life is

CDAC MUMBAI

Beautiful', 789,1992, 1800), (104, 'My Sql Foundation',963,1996,1000),(105,'Javascript', 741,1990, 900);

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> select * from Books;

```
+---------+------------------+-----------+---------------+-----------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+---------------+-----------------+
|     100 | Gitanjali        |       123 |          1961 |             100 |
|     102 | Once Upon a Time |       456 |          1962 |            1500 |
|     103 | Life is Beautiful |       789 |          1992 |            1800 |
|     104 | My Sql Foundation |       963 |          1996 |            1000 |
|     105 | Javascript       |       741 |          1990 |             900 |
+---------+------------------+-----------+---------------+-----------------+
```

5 rows in set (0.00 sec)

10. Insert 3 members into the Members table.

mysql> insert into Members(member_id, name, email, phone)values(1,'Rahul','Rahul@gmail.com',9874563210),(2,'Ram','Ram@gmail.com',984562210) ,(3,'Sham','Sham@gmail.com',987456789);

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from Members;

```
+-----------+-------+-----------------+------------+
| member_id | name  | email           | phone      |
+-----------+-------+-----------------+------------+
|         1 | Rahul | Rahul@gmail.com | 9874563210 |
|         2 | Ram   | Ram@gmail.com   | 984562210  |
```

CDAC MUMBAI

| 3 | Sham | Sham@gmail.com | 987456789 |

+----------+-------+----------------+------------+

3 rows in set (0.00 sec)

11. Insert 4 borrow records into the Borrow_Records table.

mysql> insert into Borrow_Records(record_id, member_id, book_id, borrow_date, return_date)values(201,1,456,'2026-10-01','2026-10-11'),(202,2,123,'2026-08-16','2026-09-14'),(203,3,789,'2026-07-16','2026-08-14'),(204,3,741,'2026-08-20','2026-09-16');

Query OK, 4 rows affected (0.03 sec)

Records: 4 Duplicates: 0 Warnings: 0

mysql> select * from Borrow_Records;

+----------+----------+---------+-------------+-------------+

| record_id | member_id | book_id | borrow_date | return_date |

+----------+----------+---------+-------------+-------------+

| 201 | 1 | 456 | 2026-10-01 | 2026-10-11 |

| 202 | 2 | 123 | 2026-08-16 | 2026-09-14 |

| 203 | 3 | 789 | 2026-07-16 | 2026-08-14 |

| 204 | 4 | 741 | 2026-08-20 | 2026-09-16 |

+----------+----------+---------+-------------+-------------+

4 rows in set (0.00 sec)

12. Write an SQL query to select all books where published_year is after 2015.

mysql> select * from Books;

+---------+------------------+----------+---------------+-----------------+

| book_id | title | author_id | published_year | available_copies |

+---------+------------------+----------+---------------+-----------------+

| 100 | Gitanjali | 123 | 1961 | 100 |

CDAC MUMBAI

```
|    102 | Once Upon a Time  |      456 |          1962 |            1500 |
|    103 | Life is Beautiful |      789 |          1992 |            1800 |
|    104 | My Sql Foundation |      963 |          1996 |            1000 |
|    105 | Javascript        |      741 |          1990 |             900 |
+---------+-------------------+----------+---------------+-----------------+
5 rows in set (0.00 sec)
mysql> select * from Books where published_year >2015;
Empty set (0.00 sec)
```

13. Write a SQL query to create a foreign key & primary key relationship between two tables.

```
mysql> create table Book(book_id int primary key, title varchar(20));
Query OK, 0 rows affected (0.02 sec)
mysql> create table author1(author_id int primary key, name varchar(20),book_id int,
    -> foreign key(book_id) references book(book_id));
Query OK, 0 rows affected (0.03 sec)
```

14. Write an SQL query to find all members who have borrowed the book with title 'Database Systems'.

```
mysql> select * from Books;
+---------+-------------------+----------+---------------+-----------------+
| book_id | title             | author_id | published_year | available_copies |
+---------+-------------------+----------+---------------+-----------------+
|    100 | Gitanjali         |      123 |          1961 |             100 |
|    102 | Once Upon a Time  |      456 |          1962 |            1500 |
|    103 | Life is Beautiful |      789 |          1992 |            1800 |
|    104 | My Sql Foundation |      963 |          1996 |            1000 |
```

CDAC MUMBAI

```
|    105 | Javascript       |      741 |          1990 |             900 |
+---------+------------------+-----------+---------------+-----------------+
5 rows in set (0.00 sec)
mysql> select * from books where title='Database Systems';
Empty set (0.00 sec)
```

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

```
mysql> select * from books;
+---------+------------------+-----------+---------------+-----------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+---------------+-----------------+
|    100 | Gitanjali        |      123 |          1961 |             100 |
|    102 | Once Upon a Time  |      456 |          1962 |            1500 |
|    103 | Life is Beautiful |      789 |          1992 |            1800 |
|    104 | My Sql Foundation |      963 |          1996 |            1000 |
|    105 | Javascript       |      741 |          1990 |             900 |
+---------+------------------+-----------+---------------+-----------------+
5 rows in set (0.00 sec)
mysql> update books set available_copies=available_copies-1 where book_id=105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
mysql> select * from books;
+---------+------------------+-----------+---------------+-----------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+---------------+-----------------+
|    100 | Gitanjali        |      123 |          1961 |             100 |
```

CDAC MUMBAI

```
|    102 | Once Upon a Time  |      456 |          1962 |           1500 |
|    103 | Life is Beautiful |      789 |          1992 |           1800 |
|    104 | My Sql Foundation |      963 |          1996 |           1000 |
|    105 | Javascript        |      741 |          1990 |            899 |
+---------+-------------------+----------+---------------+-----------------+
5 rows in set (0.00 sec)
```

16. Delete a record from Members where member_id = 3.

```
mysql> select * from Members;
+-----------+-------+-----------------+------------+
| member_id | name  | email           | phone      |
+-----------+-------+-----------------+------------+
|         1 | Rahul | Rahul@gmail.com | 9874563210 |
|         2 | Ram   | Ram@gmail.com   | 984562210  |
|         3 | Sham  | Sham@gmail.com  | 987456789  |
+-----------+-------+-----------------+------------+
3 rows in set (0.00 sec)
mysql> delete from Members where member_id=3;
Query OK, 1 row affected (0.01 sec)
mysql> select * from Members;
+-----------+-------+-----------------+------------+
| member_id | name  | email           | phone      |
+-----------+-------+-----------------+------------+
|         1 | Rahul | Rahul@gmail.com | 9874563210 |
|         2 | Ram   | Ram@gmail.com   | 984562210  |
+-----------+-------+-----------------+------------+
```

CDAC MUMBAI

2 rows in set (0.00 sec)

17. Update a Book name record from Book table with id = 1.

```
mysql> select * from books;
+---------+------------------+-----------+---------------+-----------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+---------------+-----------------+
|     100 | Gitanjali        |       123 |          1961 |             100 |
|     102 | Once Upon a Time  |       456 |          1962 |            1500 |
|     103 | Java             |       789 |          1992 |            1800 |
|     104 | My Sql Foundation |       963 |          1996 |            1000 |
|     105 | Javascript       |       741 |          1990 |             899 |
+---------+------------------+-----------+---------------+-----------------+
5 rows in set (0.00 sec)
mysql> update books set title='Python' where book_id=100;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
mysql> select * from books;
+---------+------------------+-----------+---------------+-----------------+
| book_id | title            | author_id | published_year | available_copies |
+---------+------------------+-----------+---------------+-----------------+
|     100 | Python           |       123 |          1961 |             100 |
|     102 | Once Upon a Time  |       456 |          1962 |            1500 |
|     103 | Java             |       789 |          1992 |            1800 |
|     104 | My Sql Foundation |       963 |          1996 |            1000 |
|     105 | Javascript       |       741 |          1990 |             899 |
```

CDAC MUMBAI

+---------+------------------+-----------+---------------+-----------------+

5 rows in set (0.00 sec)

18. Write an SQL query to list all books along with their authors' names.

mysql> select * from books;

+---------+------------------+-----------+---------------+-----------------+------------------+

| book_id | title | author_id | published_year | available_copies | author_name |

+---------+------------------+-----------+---------------+-----------------+------------------+

| 100 | Python | 123 | 1961 | 100 | Eric Matthes |

| 102 | Once Upon a Time | 456 | 1962 | 1500 | Unknown |

| 103 | Java | 789 | 1992 | 1800 | Kathy Sierra |

| 104 | My Sql Foundation | 963 | 1996 | 1000 | Unknown |

| 105 | Javascript | 741 | 1990 | 899 | Marijn Haverbeke |

+---------+------------------+-----------+---------------+-----------------+------------------+

5 rows in set (0.00 sec)

mysql> select title, author_name from books;

+------------------+-----------------+

| title | author_name |

+------------------+-----------------+

| Python | Eric Matthes |

| Once Upon a Time | Unknown |

| Java | Kathy Sierra |

| My Sql Foundation | Unknown |

| Javascript | Marijn Haverbeke |

+------------------+-----------------+

CDAC MUMBAI

5 rows in set (0.00 sec)

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

```
mysql> select * from books;
+---------+------------------+-----------+---------------+-----------------+-----------------+
| book_id | title            | author_id | published_year | available_copies | author_name     |
+---------+------------------+-----------+---------------+-----------------+-----------------+
|     100 | Python           |       123 |          1961 |             100 | Eric Matthes    |
|     102 | Once Upon a Time |       456 |          1962 |            1500 | Unknown         |
|     103 | Java             |       789 |          1992 |            1800 | Kathy Sierra    |
|     104 | My Sql Foundation |       963 |          1996 |            1000 | Unknown         |
|     105 | Javascript       |       741 |          1990 |             899 | Marijn Haverbeke |
+---------+------------------+-----------+---------------+-----------------+-----------------+
5 rows in set (0.00 sec)

mysql> delete from books where published_year < 2000;
Query OK, 5 rows affected (0.01 sec)

mysql> select * from books;
Empty set (0.00 sec)
```

20. Write an SQL query to find all books that are never borrowed (i.e., no records in Borrow_Records).

```
mysql> select * from books b
    -> where not exists
    -> (select 1 from borrow_records br where br.book_id = b.book_id);
Empty set (0.00 sec)
```

CDAC MUMBAI

CDAC MUMBAI